MATH 161, Autumn 2025
SCRIPT 5: Connectedness and Boundedness

We introduce a new axiom for a continuum C and derive many interesting properties from it. From now on, we will always assume that C satisfies axiom 4 (as well as axioms 1,2 and 3).

Axiom 4. *A continuum is connected.*

Theorem 5.1. *The only subsets of a continuum C that are both open and closed are $\emptyset$ and C.*

Proof of Theorem 5.1. We want to show that the only subset of C that are both closed and open are $\emptyset$ and C. We want to prove this through direct proof.

Suppose there exists a set A. Assume A is both open and closed and $A \neq \emptyset$ and $A \neq C$. Next, we may also say that $C \setminus A$ is also open and closed.

Now, as the decomposition is of two disjoint, nonempty, open subsets we may write C as $C = A \cup (C \setminus A)$. This contradicts **Axiom 4** which states that the a continuum is connected. Therefore, as no such subset A exists, this proves that the only subsets that may be open and closed are C and $\emptyset$. □

Corollary 5.2. *Every region is infinite.*

Proof of Corollary 5.2. We want to show that every region is infinite. We will do this through a proof by contradiction.

Suppose, for the sake of contradiction, that R is a finite region of C. Since every region is open, R is an open subset of C. Moreover, finite subsets of C are closed, and hence R is also closed.

Since R is a region, it is nonempty, and since regions are proper subsets of C, we have $R \neq C$. Thus R is a subset of C that is nonempty, not equal to C, and both open and closed. This contradicts the fact that the only subsets of a continuum that are both open and closed are $\varnothing$ and C.

Therefore, no region can be finite, and every region of C is infinite. □

Corollary 5.3. *Every point of C is a limit point of C.*

Proof of 5.3. We want to prove that every point of C is a limit point of C. We will do this through direct proof.

Suppose there exists an arbitrary value $x \in C$. As C is has a first and past point and has an ordering $<$, there exists some $a, b \in C$ such that $a < x < b$. Therefore, we may say that the region

$\underline{ab} \subset C$ and $x \in (a, b)$. Therefore, we may write $ab \cap C \setminus \{x\} \neq \emptyset$.

Now, take an arbitrary region $R \ni x$. Similarly there exists $a', b' \in C$ such that $R = (a', b')$ and $a' < x < b'$. By the same argument we see $R \cap C \setminus \{x\} \neq \emptyset$ and therefore every region containing x contains a point in C distinct from x. Hence, by the properties of the continuum as taking arbitrary values of x, we may say every point of C is a limit point of C. $\square$

Corollary 5.4. *Every point of the region $\underline{ab}$ is a limit point of $\underline{ab}$.*

Proof of Corollary 5.4. Let $p \in \underline{ab}$ be arbitrary. Since $a < p < b$ and the continuum has no first or last point, there exist elements $c, d \in C$ such that

$$a < c < p < d < b.$$

Then the region $\underline{cd}$ satisfies

$$p \in \underline{cd} \subset \underline{ab}.$$

By Corollary 5.2, every region is infinite, and hence

$$\underline{cd} \cap (\underline{ab} \setminus \{p\}) \neq \emptyset.$$

Now let R be an arbitrary region containing p. Then $R = \underline{ef}$ for some $e, f \in C$ with $e < p < f$. By the same argument as above, R contains a region $\underline{cd}$ with

$$p \in \underline{cd} \subset R \subset \underline{ab},$$

and therefore

$$R \cap (\underline{ab} \setminus \{p\}) \neq \emptyset.$$

Thus every region containing p contains a point of $\underline{ab}$ distinct from p, so p is a limit point of $\underline{ab}$. Since p was arbitrary, every point of $\underline{ab}$ is a limit point of $\underline{ab}$. $\square$

We will now introduce boundedness. The first definition should be intuitively clear. The second is subtle and powerful. Note that for these definitions and Exercises 5.7-5.9 we only need the continuum to satisfy Axioms 1-3.

Definition 5.5. Let X be a subset of C. A point u is called an *upper bound* of X if for all $x \in X$, $x \leq u$. A point l is called a *lower bound* of X if for all $x \in X$, $l \leq x$. If there exists an upper bound of X, then we say that X is *bounded above.* If there exists a lower bound of X, then we say that X is *bounded below.* If X is bounded above and below, then we simply say that X is *bounded.*

Definition 5.6. Let X be a subset of C. We say that u is a *least upper bound* of X and write $u = \sup X$ if:

1. u is an upper bound of X, and
2. if u' is an upper bound of X, then $u \leq u'$.

We say that l is a *greatest lower bound* and write $l = \inf X$ if:

1. l is a lower bound of X, and
2. if l' is a lower bound of X, then $l' \leq l$.

The notation sup comes from the word *supremum*, which is another name for least upper bound. The notation inf comes from the word *infimum*, which is another name for greatest lower bound.

Exercise 5.7. If $\sup X$ exists, then it is unique, and similarly for $\inf X$.

Exercise 5.7. We want to show that if sup X exists, then it is a unique value. We will show this by a proof by contradiction.

Suppose there exists sup X. Let sup $X = u$. For the same of contradiction, suppose there exists some other supremum u' such that $u' \neq u$. Therefore it may be that $u' < u$ or $u' > u$.

As $u =$ sup X, and by **Definition 5.6**, we know that $u' < u$ is not true as we know u is the lowest upper bound. If $u' > u$, then u' is not the lowest upper bound. Therefore, it must be that $u' = u$ and therefore sup u is unique.

An analogous argument shows that if $\inf X$ exists, then it is unique. □

Exercise 5.8. If X has a first point L, then $\inf X$ exists and equals L. Similarly, if X has a last point U, then $\sup X$ exists and equals U.

Exercise 5.8. Suppose there exists set X which contains first point L and last point U. We want to show that inf $X = L$ and sup $X = U$.

By the definition of a first point, we know that there does not exist any $x \in X$ such that $x < L$. Therefore, we may say $\forall x \in X, L \leq x$. This may be written as $\forall x \in X, L \leq x$. Hence, we may say that L is a lower bound.

Now, let l' be a lower bound. If l' is a lower bound, $\forall x \in X, l' \leq x$ and, as the infimum is unique, we may say $l' < L$. Hence, by **Definition 5.6**, we can say L is the greatest lower bound.

The proof that $\sup X = U$ when X has a last point U is analogous. Thus, this proves that the first and last point of a set equal the infimum and supremum respectively. □

Exercise 5.9. For this exercise, we assume that $C = \mathbb{R}$. Find $\sup X$ and $\inf X$ for each of the following subsets of $\mathbb{R}$, or state that they do not exist. You need not give proofs.

1. $X = \mathbb{N}$

   Answer for part 1. inf $X = 1$ and sup X does not exists as N is unbounded above. □

2. $X = \mathbb{Q}$

   Answer for part 2. $\inf X$ and $\sup X$ does not exists as $\mathbb{Q}$ is unbounded below and unbounded above. $\square$

3. $X = \{\frac{1}{n} \mid n \in \mathbb{N}\}$

   Answer for part 3. $\inf X = 0$ as $\frac{1}{n}$ approaches 0 as n approaches infinity. $\sup X = 1$. $\square$

4. $X = \{x \in \mathbb{R} \mid 0 < x < 1\}$

   Answer for part 4. $\inf X = 0$ and $\sup X = 1$. $\square$

5. $X = \{3\} \cup \{x \in \mathbb{R} \mid -7 \leq x \leq -5\}$

   Answer for part 5. $\inf X = -7$ and $\sup X = 3$. $\square$

The following lemma is extremely useful when dealing with suprema; an analogous statement can be made for infima.

Lemma 5.10. *Let C satisfy Axioms 1-3. Suppose that X is a subset of C and $s = \sup X$ exists. If $p < s$, then there exists an $x \in X$ such that $p < x \leq s$.*

Proof of Lemma 5.10. Suppose $s = \sup X$ exists. Now suppose, for the sake of contradiction, that there exists $p < s$ such that no $x \in X$ satisfies $p < x \leq s$. Then every $x \in X$ satisfies $x \leq p$, and hence p is an upper bound of X. This contradicts the definition of $s = \sup X$ as the least upper bound, since $p < s$ and p is also an upper bound. Therefore, there must exist $x \in X$ such that $p < x \leq s$. $\square$

Theorem 5.11. *Let C satisfy Axioms 1-3, but not necessarily Axiom 4. Suppose also that all regions of C are nonempty. Let $a < b$. The least upper bound and greatest lower bound of the region $\underline{ab}$ are:*

$$\sup \underline{ab} = b \quad \textit{and} \quad \inf \underline{ab} = a.$$

Proof of 5.11. We want to show that $\sup \underline{ab} = b$ and $\inf \underline{ab} = a$.

Firstly, we must show that for region $\underline{ab}$, b is an upper bound and a is a lower bound. As $\underline{ab} \subset C$, we may say that $\forall x \in \underline{ab},\ a < x < b$. Therefore, we may say that

$$\forall x \in X,\ x < b \text{ and } \forall x \in X,\ x > a$$

This may be written as

$$\forall x \in X,\ x \leq b \text{ and } \forall x \in X,\ x \geq a$$

Hence, by **Definition 5.5** we may say that a and b are lower and upper bounds respectively.

Now, we show that $\inf \underline{ab} = a$ and $\sup \underline{ab} = b$. Let c be an arbitrary lower bound of $\underline{ab}$. Then by definition, $c \leq x$ for all $x \in \underline{ab}$. Since every $x \in \underline{ab}$ satisfies $a < x$, we have $c \leq x$ for all $x > a$, which implies $c \leq a$. Therefore, a is greater than or equal to every lower bound, and hence $a = \inf \underline{ab}$.

Similarly, let d be an arbitrary upper bound of $\underline{ab}$. Since every $x \in \underline{ab}$ satisfies $x < b$, we have $d \geq x$ for all $x < b$, which implies $d \geq b$. Hence, $b = \sup \underline{ab}$.

This proves that sup $\underline{ab} = b$ and inf $\underline{ab} = a$. □

Lemma 5.12. *Let C satisfy Axioms 1-3. Let X be a subset of C and suppose* $\sup X$ *exists. Then* $\sup X \in \overline{X}$. *Similarly,* $\inf X \in \overline{X}$.

Proof of 5.12. We want to show that for any subset X of C, we want to show sup $X \in \overline{X}$ and inf $X \in \overline{X}$. For this, it suffices to show that the supremum and the infin um are either in X or the limit points of X.

Consider two cases: set X is finite and set X is infinite.

Case 1: Set X is finite Suppose that X is finite. Then, we may say there exists a first point a and a last point b. By **Exercise 5.8**, we may say that inf $X = a$ and sup $X = b$. As the first and last point are in X, we may say that sup $X \in X$ and inf $X \in X$. Thus,sup $X \in \overline{X}$ and inf $X \in \overline{X}$

Case 2: Set X is infinite Suppose that X is infinite and that $\sup X$ and $\inf X$ exist. If $\sup X \in X$, then trivially $\sup X \in \overline{X}$. Otherwise, if $\sup X \notin X$, and we need to show that it is a limit point of X. Let R be any region containing $\sup X$. By the definition of supremum and **Lemma 5.10**, for every $p < \sup X$, there exists $x \in X$ such that $p < x \leq \sup X$. Since R contains $\sup X$, it also contains some $p < \sup X$, so there exists $x \in R \cap X$ with $x \neq \sup X$. Hence $R \cap (X \setminus \{\sup X\}) \neq \emptyset$, and therefore $\sup X \in LP(X)$.

By a completely analogous argument, considering lower bounds instead of upper bounds, we can similarly conclude that $\inf X \in \overline{X}$. □

Corollary 5.13. *Let C satisfy Axioms 1-3, but not necessarily Axiom 4. Suppose also that all regions of C are nonempty. Then both a and b are limit points of the region $\underline{ab}$.*

Proof of Corollary 5.13. We want to show that a and b are limit points of the region $\underline{ab}$.

By **Theorem 5.11**, we know that for a nonempty region of C that inf $\underline{ab} = a$ and sup $\underline{ab} = b$. Furthermore, by **Corollary 5.12**, we know that the supremum and infimum are in the closure of $\underline{ab}$. Therefore, sup $\underline{ab}$, inf $\underline{ab} \in \underline{ab} \cup LP(\underline{ab})$. Then, since $\underline{ab}$ is infinite, we know that sup $\underline{ab}$ and inf $\underline{ab}$ are not in $\underline{ab}$. Hence, we may say that sup $\underline{ab}$ and inf $\underline{ab}$ are limits points of the region $\underline{ab}$.

Hence this works to prove that a and b are the limit points of region $\underline{ab}$.

□

Let $[a, b]$ denote the closure $\overline{\underline{ab}}$ of the region $\underline{ab}$.

Corollary 5.14. *Let C satisfy Axioms 1-3, but not necessarily Axiom 4. Suppose also that all regions of C are nonempty. Then $[a, b] = \{x \in C \mid a \leq x \leq b\}$.*

Proof of Corollary 5.14. We want to show that all if all regions of C are nonempty then $[a, b] = \{x \in C \mid a \leq x \leq b\}$. We may write $[a, b]$ as $\underline{ab} \cup LP(\underline{ab})$. This means that $\forall x \in [a, b]$ we may say that $x \in \underline{ab}$ or $x \in LP(\underline{ab})$.

We know that for any $x \in [a, b]$, either $x \in \underline{ab}$ or $x \in LP(\underline{ab})$. If $x \in \underline{ab}$, then by the definition of a region we have $a < x < b$. If $x \in LP(\underline{ab})$, then by **Corollary 5.13** the limit points of $\underline{ab}$ are a and b, so $x = a$ or $x = b$. Combining these cases, we conclude that for all $x \in [a, b]$, $a \leq x \leq b$, as desired. $\square$

Lemma 5.15. *Let C satisfy Axioms 1-3, $X \subset C$ and define:*

$$\Psi(X) = \{x \in C \mid x \textit{ is not an upper bound of } X\}.$$

Then $\Psi(X)$ is open. Define:

$$\Omega(X) = \{x \in C \mid x \textit{ is not a lower bound of } X\}.$$

Then $\Omega(X)$ is open.

Proof of 5.15. Let $x \in \Psi(X)$. We want to show that $\Psi(X)$ is open. Let there be some y such that $y > x$. Then, we may construct a region $\underline{ay}$ such that $a < x < y$. Let $m \in \underline{ay}$ such that $a < m < y$. To show that $\Psi(X)$ is open it suffices to prove that region $\underline{ay} \subset \Psi(X)$.

Since $x \in \Psi(X)$, there exists $n \in X$ with $n > x$. For any $m \in \underline{ay}$, we may choose the region such that $n > m$ so $m \in \Psi(x)$ Therefore, we may say that m is not an upper bound of X and hence $m \in \Psi(X)$. Thus, as m is an arbitrary element in $\underline{ay}$ we may say that this applies for all elements. Therefore $\underline{ay} \subset \Psi(X)$ and $\Psi(X)$ is an open set.

A completely analogous arguement show that $\Omega(X)$ is open.

$\square$

Theorem 5.16. *Suppose that C satisfies Axioms 1-4 and $X \subset C$ is nonempty and bounded above. Then $\sup X$ exists. Similarly, if X is nonempty and bounded below, then $\inf X$ exists.*

Proof of Theorem 5.16. Let $X \subset C$ be a nonempty subset of C that is bounded above. We want to show that sup X exists through a proof by contradiction.

Construct the sets

$$U = \{y \in C \mid y \text{ is an upper bound for } X\}$$

and

$$L = \{y \in C \mid y \text{ is not an upper bound of } X\}$$

As X is bounded above we may say that U is nonempty. As C has no first point, there exists an element of C that is not an upper bound of X and therefore L is nonempty. By construction, we may also say these sets are disjoint and $C = L \cup U$. Furthermore, every element in L is less than every element of U.

For the sake of contradiction, suppose that $\sup X$ does not exist. Then U has no least element. Furthermore, for every $u \in U$ there exists $u' \in U$ such that $u' < u$. This implies that there is no element of C that separates L and U. We may alternatively state this as there is no $c \in C$ such that every element of L is less than or equal to c and every element of U is greater than or equal to c.

This C is split into two non empty sets L and U such that every element of L is strictly less than every element of U., and there is no element of C lying between them. This is a contradiction of axiom 4. Therefore, as there exists a contradiction, U must have a least element which is the least upper bound of X.

A completely analogous argument applies for $\inf X$. □

Corollary 5.17. *Suppose that C satisfies Axioms 1-4. Then every nonempty closed and bounded set has a first point and a last point.*

Proof of Corollary 5.17. Suppose there exists set X which is nonempty, closed, and bounded. We want to show that the set has a first point and a last point.

By the definition of X, we may say that there exists a $\sup X$ and an $\inf X$ as the set is bounded. Furthermore, we may say $X = \overline{X}$. By **Lemma 5.12** we know that $\sup X \in \overline{X}$ and $\inf X \in \overline{X}$. Therefore, as $X = \overline{X}$, we may say that $\sup X \in X$ and $\inf X \in X$. Since, by the work in **Exercise 5.8**, we know that $\sup X$ is the last point and $\inf X$ is the first point. Therefore, we may say that the first and last points are in X. This works to prove that every nonempty closed and bounded set has a first and a last point. □

Exercise 5.18. Is this true for $\mathbb{Q}$? More explicitly, is it true that in $\mathbb{Q}$, every nonempty closed and bounded set has a first point and a last point?

Exercise 5.18. We want to show that this statement is false.

Consider the set
$$X = \{q \in \mathbb{Q} \mid q^2 < 2\}.$$
This set is nonempty, since $1 \in X$, and it is bounded above, by for example 2, and bounded below, by for example 0. Suppose, for the sake of contradiction, that X has a first point $q_{\min} \in X$. Then $q_{\min}^2 < 2$. Now, choose a natural number N large enough so that $q = q_{\min} - \frac{1}{N} > 0$ and $q^2 < 2$.

Therefore, we may say that $q \in X$ and $q < q_{\min}$. This contradicts the assumption that $q_{\min}$ is the first point. Therefore, X has no first point.

Similarly, suppose for the sake of contradiction that X has a last point $q_{\max} \in X$. Then $q_{\max}^2 < 2$, and we can choose N large enough so that $q = q_{\max} + \frac{1}{N}$ satisfies $q^2 < 2$.

Therefore, we may say that $q \in X$ and $q > q_{\max}$. This contradicts the assumption that $q_{\max}$ is the last point. Hence X has no last point either.

Thus, X is nonempty, bounded, and closed in $\mathbb{Q}$, yet it has neither a first point nor a last point. This provides a counterexample showing that in $\mathbb{Q}$, it is not true that every nonempty closed and bounded set has a first or last point.

$\square$

Additional Exercises

1. For this exercise, we assume that $C = \mathbb{R}$. Find $\sup X$ and $\inf X$ for each of the following subsets of $\mathbb{R}$, or state that they do not exist. You should give proofs.

   (a) $X = \{-2^n \mid n \in \mathbb{N}\}$

   (b) $X = \{x \mid a \leq x \leq b\}$

   (c) $X = \{-1\} \cup \{\frac{1}{n} \mid n \in \mathbb{N}\}$

   (d) $X = \bigcup_{k \geq 0} \{\frac{k}{n} \mid n \in \mathbb{N}\}$.

2. Prove that if A, B are subsets of a continuum C and $a \leq b$, for all $a \in A, b \in B$, then

$$\sup A \leq \inf B.$$

3. Let $a, b \in C$ with $a < b$. Show that $\underline{ab}$ is a continuum.

4. Assume that C satisfies Axioms 1,2 and 3. Do not assume that C satisfies Axiom 4. Suppose also that

   (a) every nonempty subset X of C that is bounded above has a supremum

   (b) all regions of C are nonempty.

   Show that C is connected.

   Hint: Argue by contradiction.